

Week 2 Assignment

Repository Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone>

Report Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone/blob/main/notebooks/main.ipynb>

Guidelines:

- 10-page max, 1.5-spaced paragraphs, 11-sized font, a professional font format. An unlimited number of appendix pages. Submitted as pdf within your GitHub repository

Sections:

1. What is the target variable and why?

Our target variable is status, which categorizes each startup's outcome as one of four values: operating, acquired, closed, or ipo (i.e., went public). We chose status as the target because our project's goal is to predict startup outcomes beyond simply looking at total funding raised. A total-funding-only view treats two startups that raised the same amount over very different timelines as equivalent, even though the path each company took to reach that funding level tells a very different story. By using status as the outcome we're predicting, we can incorporate funding velocity and timing patterns (how quickly rounds occurred relative to founding and to each other) as predictors, since those signals say more about a company's trajectory than the funding total alone.

2. What are the predictors and why?

The predictors within our project are the funding velocity and the funding cadences. The idea of emphasising the funding velocity and cadences as our predictors is to take away the importance of total funding raised as our key output. The key features within our project include the amount of time it takes to obtain the first interval of funding, funding speed, and funding round frequency. These 3 features calculate momentum of the startup signaling success or failure. A company that is able to secure three rounds of funding within 2 years shows that positive scaling as well as investor demand. If we raise that total over a 10 year cadence, this will portray a very different story of slow steady growth or the startup has become stagnant.

Alongside timing, we decided to use simple baseline predictors such as industry sectors, location, and average cash per round. The importance of these baseline predictors shows us how different industries operate on a regulatory level. For example, a biotechnology startup requires much more time and funding to reach an exit in comparison to a software startup. Additionally, location plays a crucial role in funding consistency and networks. Considering these everyday background factors with our prediction metrics will give our model a clear and complete picture of both the startup's timing and the cadence of its outer environment.

These predictors work together to help our model distinguish between all four of our outcomes rather than treating overall success as a binary metric. Funding velocity combined with high capital per round shows a strong correlation with high value outcomes such as an IPO or acquisition. However, long lags between rounds, paired with low capital cadence often signals that a startup is struggling. By capturing both predictors, the features that we produce will be able to give our model the exact parameters it needs to separate a fast growing startup from a stagnant one.

3. Exploration of the dataset: definition of variables, data types, general dataset stats: count of rows, count of columns, etc.

The dataset we're using has a total of 66,368 rows and 14 columns, which gives us a decent amount of startup data to work with. Most of the columns are objects (strings), with `funding_rounds` being the only "true" numeric column at the moment. The variable `funding_total_usd` should be numeric, but it's currently stored as a string. It is only happening because of the "-" characters filling in missing values, so it's something that can be easily cleaned up before we use it as a predictor.

As for missing values in our dataset, the column `founded_at`, which tells us when a startup was founded, is missing for nearly 23% of rows. This is a problem since it's a key piece of the funding velocity/timing story we're trying to tell with our target variable. There are plenty of different techniques for dealing with this, and working through this problem should give us good practice handling real-life, messy data. `State_code`, `region`, and `city` are also missing at similar rates (about 12-13%), which makes sense since those would mostly be missing together for the same international/unlisted companies.

For our target variable, status is heavily imbalanced. Nearly 80% of the startups in the dataset are still operating, with closed and acquired each making up under 10% and ipo only around 2.3%. This will matter a lot once we get to modeling, and will give us good experience working with techniques like resampling and class weights later in the project.

Overall, this dataset gives us real-world experience dealing with imperfect data, which will help us build stronger data cleaning and modeling skills as the project progresses.